

NIRVANA – Master Specification

Volume 7: Data Engine & Provider Layer

Objective

Create a modular data acquisition system that separates providers from business logic.

Data Providers

Prototype providers: Investing.com (market data), Paisabazaar (credit cards), Excel spreadsheets (master editable data), with cached fallback datasets.

Update Data Engine

Admin button triggers synchronization of stocks, ETFs, mutual fund NAVs, gold, bond yields, FD rates, macro indicators, tax rules and credit-card offers before rebuilding the normalized database.

Normalization Pipeline

Fetch → Validate → Clean → Standardize → Deduplicate → Categorize → Version → Persist → Notify Recommendation Engine.

Excel Integration

Support upload, schema validation, preview, conflict detection, rollback, version history and export of normalized datasets.

Provider Health

Monitor sync status, latency, failures, last update timestamp and record counts. Automatically switch to cached Excel data if live synchronization fails.

Audit & Logging

Maintain logs for imports, updates, failures, manual edits, provider changes and recommendation data snapshots for reproducibility.

Future Expansion

Architecture should support replacement of prototype providers with official APIs without changing downstream recommendation logic.